Fall26_SLS775. Natural Language Processing in Corpus Research

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Contents

1 Introduction to NLP for corpus linguistics

References

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Corpus linguistics and natural language processing (NLP) both investigate language through collections of naturally occurring linguistic data. However, the two fields have traditionally differed in their primary goals. Corpus linguistics generally begins with a linguistic question and uses corpus data to describe patterns of language use, while NLP develops computational methods for automatically analyzing or generating human language. NLP for corpus linguistics brings these perspectives together: computational models are used to identify, annotate, measure, and compare linguistic patterns in corpora at a scale that would difficult to achieve through manual analysis alone.

This combination has become increasingly important as the size and diversity of available language data have expanded. Early electronic corpora, such as the Brown Corpus, contained approximately one million words, whereas contemporary corpora may contain billions of words. Such datasets create new opportunities for investigating linguistic variation, but they also make exclusively manual analysis impractical. Computational methods therefore allow corpus linguists to scale up their analyses while maintaining an explicit and, in principle, reproducible analytical procedure.

One way to organize computational problems in corpus linguistics is to distinguish between two broad types of analysis: (1) *categorization* and (2) *comparison*. Categorization involves assigning a predefined label to a linguistic unit. For example, a corpus linguist may classify a word by its part of speech, determine whether a sentence instantiates a particular grammatical construction, or identify the genre to which a document belongs. Comparison, by contrast, involves assessing the degree of similarity or relatedness between linguistic units. A researcher may examine whether two words are semantically related, whether two sentences convey similar meanings, or whether two texts share stylistic characteristics.

These two types of analysis are not rigidly tied to particular machine-learning algorithms. Note that in *supervised* learning, a model is trained on examples that have already been assigned *target labels* (in other words, annotated tags). The model learns the relationship between the input features and the expected output and then applies that relationship to new data. In *unsupervised* learning, by contrast, the data are not provided with predefined target labels. The model instead identifies patterns, groupings, or relationships within the data.

Categorization is often implemented through supervised learning because researchers typically define the categories that a model is expected to predict. For example, a researcher may provide texts labeled as academic writing, news, or conversation and train a classifier to assign new texts to one of these registers. However, categories can also be explored through unsupervised methods. A clustering algorithm might group texts according to their linguistic similarities, after which the researcher examines whether the resulting clusters correspond to recognizable registers.

Comparison is often conducted using representations or similarity measures that do not require predefined class labels. For example, a researcher may represent two sentences as vectors and calculate their cosine similarity to estimate how closely related their meanings are. However, comparison can also involve supervised learning. A model may be trained on sentence pairs that have been labeled with degrees of semantic similarity and then used to predict the similarity of new sentence pairs.

1.1 What is corpus linguistics?

Corpus linguistics is an *empirical* approach to the study of language based on the systematic analysis of collections of linguistic data known as *corpora*. At the most basic technical level, a corpus may consist of multiple machine-readable text files accompanied by information about the texts, speakers, and contexts from which the data were collected (i.e., metadata). However, a corpus is not simply any large collection of texts. Its value as a research resource depends on how systematically the data were selected, organized, and documented. This distinction also helps clarify the difference between linguistic *data* and a *corpus*: whereas data may refer broadly to any collection of linguistic observations, a corpus is a deliberately constructed and documented dataset designed to support systematic analysis of a particular population or domain of language use.

A corpus is generally designed to represent a particular population or domain of language use, such as a group of speakers, a register, a genre, a topic, a historical period, or a language variety. Its sampling scheme should therefore reflect relevant sources of variability within the population it is intended to represent. For example, a corpus of contemporary spoken English may sample speakers from different regions, age groups, and communicative settings. Claims based on a corpus are consequently limited by the population and contexts represented in its design. We will return to this issue in our discussion of corpus *representativeness*.

Corpus linguistics is primarily concerned with the distribution of linguistic forms and functions. Researchers examine how frequently linguistic elements occur, the contexts in which they occur, the other elements with which they co-occur, and how these distributions vary across speakers, texts, registers, or time periods. These distributional patterns can be summarized through measures (e.g., token frequency, association strength).

Corpora are commonly used in psycholinguistics as sources of quantitative measures derived from the distributional patterns of language use. These measures can serve as operationalizations of psycholinguistic constructs that are not directly obserable. In other words, corpus statistics may not be able to measure cognitive representations or processes directly, but they quantify aspects of an individual's likely linguistic experience that may be relevant to how language is (learned and) used. For example, one of the most basic corpus measures is *token frequency*, or the number of times a particular linguistic form or pattern occurs. Token frequency can be related to *entrenchment*, the process through which repeated exposure strengthens a linguistic representation and allows it to become established as a cognitive routine.[1] Similarly, the frequency with which two linguistic elements occur together may contribute to the entrenchment of that combination. Corpus data also make it possible to quantify *contingency*, or the extent to which the occurrence of one linguistic element predicts the occurrence of another. Such relationships can be investigated through analyses of collocation and/or collostruction, often using conditional probabilities or association measures. For example, researchers may examine whether a particular verb occurs with a given construction more frequently than would be expected from the independent frequencies of the verb and the construction. Strong contingency may facilitate the learning of form–form or form–meaning associations because the occurrence of one element provides reliable information about another.

One important advantage of corpus data is their potential for ecological validity.[2] Experimental studies are often conducted under highly controlled conditions using carefully selected and sometimes decontextualized stimuli. Such control is necessary for establishing causal relationships, but the resulting language may not always reflect the distributions encountered in ordinary communication. Balanced experimental designs may also expose participants to linguistic forms in proportions that differ substantially from those found in everyday language use. Corpus data complement experimental evidence by documenting how language is distributed in naturally occurring, less experimentally constrained communicative contexts.

At the same time, corpus data present several analytical challenges. Naturally occurring language is often noisy and heterogeneous, and linguistic variables are frequently highly intercorrelated. Contexts may vary considerably both within and across speakers, making it difficult to isolate the influence of any single factor. Corpus distributions are also typically highly imbalanced and approximately Zipfian: a small number of linguistic elements occur very frequently, whereas most occur only rarely. Consequently, corpus frequencies and associations must be interpreted in light of corpus composition, sampling decisions, dispersion, and the number of relevant opportunities for occurrence.

[1] For example, a word, phrase, or grammatical construction that occurs frequently in language use may be recognized or processed more readily than a less frequently encountered alternative.

[2] Ecological validity refers to "the degree to which a given research context is indicative of, or can be generalized to, real-world environments" (Schmuckler, 2026). More details here: `https://oecs.mit.edu/pub/1y8kpgak`

Corpus linguistics is therefore not simply the automated counting of words in large text collections. It is a research approach that connects systematically sampled language data, quantitative descriptions of linguistic distributions, and theoretically motivated interpretations of language use. Its strength lies in allowing researchers to investigate how linguistic patterns emerge across large numbers of contextualized observations while remaining attentive to the limitations of the data and the procedures used to analyze them.

1.2 What is NLP?

NLP is concerned with the automated analysis and generation of human language. NLP systems process written or spoken language in order to identify linguistic properties, extract information, make predictions, or produce new language. Common NLP tasks include dividing texts into words and sentences, assigning part-of-speech (POS) categories, identifying syntactic relationships, measuring semantic similarity, translating between languages, classifying documents, answering questions, and generating text. Speech recognition and speech synthesis are also closely related to NLP, although speech processing is sometimes treated as a separate subfield.

NLP can operate at different levels of linguistic analysis. At the word level, a system may identify lemmas, morphological features, or POS tags. At the sentence level, it may analyze grammatical structure, identify errors, or represent meaning. At the text level, it may measure cohesion, classify genre, estimate proficiency, or generate a summary. NLP therefore provides methods for transforming unstructured language into representations that can be systematically searched, counted, compared, and modeled.

This role makes NLP particularly important for corpus linguistics. A corpus may contain thousands or millions of texts that cannot be analyzed manually in their entirety. NLP tools can automatically annotate these texts and extract linguistic features relevant to a research question. For example, a researcher may use an NLP pipeline to identify verbs and their subjects, calculate lexical and syntactic complexity measures for individual texts, or represent texts as vectors for classification and comparison. NLP therefore supports the scalability of corpus analysis. However, the validity of the resulting analysis depends on several factors: the capabilities and limitations of the computational method used (e.g., a simple count-based system vs. Transformer architecture), the characteristics of the corpus to which it is applied (e.g., edited standard-language texts from news sources vs. L2 learner language), and the extent to which the computational output provides a valid operationalization of the linguistic construct under investigation (e.g., text length vs. grammatical complexity).

In the context of language learning, NLP has two broad applications. The first is the analysis of *learner language*: words, sentences, and texts produced by language learners. The second is the analysis of *language for learners*: texts and other language materials that are selected, adapted, or generated to support learning.

The first broad application includes areas such as intelligent language tutoring systems, automatic language assessment, and grammatical error detection. An intelligent language tutoring system may analyze a learner's response in order to provide individualized feedback, select an appropriate subsequent activity, or update a model of the learner's developing abilities. NLP is necessary in such systems because it is generally impossible to anticipate every response that a learner might produce and manually specify the appropriate feedback for each possible string. Instead, the system must abstract away from the exact wording of the response and identify more general properties of its form, meaning, and appropriateness. This requirement creates a distinctive challenge for NLP. Many general-purpose NLP systems are designed to remain robust when input contains errors/unexpected forms. A parser, for instance, is typically expected to produce a syntactic analysis even when a sentence is not fully well formed. In a language tutoring system, however, the unexpected form may itself be the information that the system needs to identify. The system must determine not only that a learner' s response differs from an expected target but also what kind of difference has occurred, what it may reveal about the learner' s developing language system, and what feedback would help the learner understand the problem. High correction accuracy alone is therefore not always sufficient. A system that produces an acceptable corrected sentence without identifying the linguistic source of the problem may provide limited insight into the learner' s underlying difficulty and limited support for learning.

NLP is also used to annotate and analyze learner corpora with a goal to investigate broader patterns of development and variation. NLP can support error annotation, identify overuse/underuse of linguistic forms, calculate measures of complexity, accuracy, and fluency, and examine relationships between linguistic features and proficiency. It can also help researchers investigate cross-linguistic influence by identifying patterns associated with learners' first-language backgrounds. By automating at least part of this process, NLP makes it possible to analyze larger learner corpora and connect theoretical questions more directly to relevant instances in the data. This application closely reflects the role of NLP in corpus linguistics introduced in Section 1 and will be a central focus throughout this course.

The second broad application involves analyzing language that is presented to learners. For example, NLP can help identify reading materials that contain particular vocabulary or grammatical structures, estimate the difficulty or readability of texts, highlight target forms, provide links to definitions or grammatical explanations, and generate exercises or assessment items from existing materials. Such applications can give learners more individualized access to language resources and allow instructional materials to be selected according to their proficiency, goals, or current learning needs.

We briefly discussed NLP within the context of applied linguistics, but the field can be defined more broadly when viewed from the perspectives of computer science and artificial intelligence. The following definition, adapted from Vajjala et al. (2020, p. 6), highlights this broader perspective:

NLP is an interdisciplinary field at the intersection of computer science, artificial intelligence, and linguistics. It focuses on developing computational systems that can process, analyze, and understand human language. Although NLP was historically concentrated in academic institutions and research laboratories, recent advances have enabled its widespread adoption in areas such as retail, healthcare, finance, law, marketing, and human resources. Several developments have contributed to this expansion:

1. the growing availability of accessible NLP tools, techniques, and application programming interfaces (APIs);
2. improvements in generalizable and interpretable approaches to complex tasks, including open-domain dialogue and question answering;
3. increased investment in interactive consumer technologies that use language as a primary means of communication;
4. the availability of open-source datasets and standardized benchmarks; and
5. the development of datasets and language-specific models for languages beyond English and other widely digitized languages.

Chapter Summary

Corpus linguistics and NLP both use collections of language data, but they have traditionally emphasized different goals. Corpus linguistics begins with a question about language use and examines the distribution of linguistic forms and functions in (systematically constructed) corpora. NLP develops computational methods for processing, analyzing, and generating language. NLP for corpus linguistics brings these perspectives together by using computational methods to annotate, search, measure, categorize, and compare linguistic patterns across datasets that may be too large for entirely manual analysis.

Activity: Brainstrom your project for this course

You might not have a concrete research question yet. Start with a topic that you are currently studying or may be interested in studying throughout the course.

1. Briefly describe your research area or a research question that interests you. What aspect of language, language use, or language learning would you like to investigate?
2. What kind of language data could help you investigate this question? For example, would you examine learner writing, classroom interaction, interviews, social media posts, published texts, assessment responses, or another type of data?
3. Could this dataset be considered a corpus? What information would you need about its speakers, writers, texts, tasks, or contexts in order to interpret it appropriately?
4. What linguistic feature, pattern, or construct would you examine in the data? Examples might include vocabulary use, grammatical constructions, errors, syntactic complexity, semantic similarity, interactional features, or differences across learner groups.
5. Would your analysis mainly involve *categorization*, *comparison*, or both? Explain your choice using one example from your research topic.
6. What part of the analysis might NLP help automate? What would you still want a researcher to examine or verify manually?
7. Identify potential challenge(s). For example, could the corpus be unrepresentative, could the NLP tool perform poorly on the language data, or could the computational measure fail to capture the intended linguistic construct?

Group report: Select one research example from your group and briefly explain (1) the research question, (2) the proposed language data, and (3) one way that NLP could support the analysis.

2 NLP in learner corpus research

References

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2.1 What is learner corpus research?

Learner corpus research (LCR) is a field that uses learner corpora to investigate the characteristics and development of learner language. Learner corpus research has generally pursued two closely connected objectives. The first is *theoretical*: LCR seeks to contribute to second language acquisition (SLA) research by providing more detailed descriptions of learner language and by improving our understanding of the factors that influence it. These factors may include learners' first-language backgrounds, proficiency levels, learning contexts, task characteristics, and cognitive or affective variables. The second objective is *pedagogical*: findings from learner corpora can be used to develop teaching materials, assessment practices, and language-learning tools that more accurately address learners' needs.

2.1.1 Learner corpus data

Early learner corpora focused predominantly on L2 English. Since then, learner corpora have been developed for a much wider range of target languages. An overview of existing resources is available through the Learner Corpora around the World[3] database. Nevertheless, English remains disproportionately represented, particularly in corpora of academic writing.

A basic distinction can be made between *raw* and *annotated* learner corpora. Raw learner corpora contain textual or transcribed spoken data in their original, relatively unprocessed form (usually in `.txt` format). They may include learner metadata (e.g., first language, age, proficiency, instructional context, or task type - usually in a separated file), but the linguistic data themselves have not been systematically enriched with additional analytical information. Annotated learner corpora include information about particular linguistic features (could be in `.xlm` or `.conllu`[4] formats). Depending on the research purpose, annotation may cover parts of speech, morphological structure, syntactic dependencies, semantic categories, pragmatic functions, discourse features, pronunciation, disfluencies, or characteristics of spoken interaction. Some forms of annotation can be produced automatically through tools such as part-of-speech taggers, morphological analyz-

[3] https://uclouvain.be/en/research-institutes/ilc/cecl/learner-corpora-around-the-world.html
[4] https://universaldependencies.org/format.html

ers, and syntactic parsers. When these tools perform with sufficiently high accuracy, they make it possible to process much larger collections of learner language than would be feasible through entirely manual analysis.

Other forms of annotation continue to require substantial human judgment. Error annotation is an important example. Researchers may identify a problematic form, categorize the type of error, and sometimes reconstruct the learner's intended target form. However, both error detection and error categorization are inherently interpretive. An annotator must decide whether an unusual form is actually erroneous, what the learner may have intended, and which linguistic category best describes the problem. These judgments may be especially difficult when a sentence contains several interacting errors. For this reason, manually annotated learner corpora should be accompanied by detailed annotation manuals that define the categories and provide examples of difficult cases. Researchers should also report intra-rater or inter-rater reliability statistics when the annotation depends on human judgment. Reliability assessment does not eliminate subjectivity, but it makes the annotation process more transparent and provides evidence regarding the consistency of the resulting data.

2.1.2 Methodological approaches and research designs

The characteristics of learner corpus data have led to the development of methodological approaches specifically designed for investigating learner language. One of the best-known approaches is *Contrastive Interlanguage Analysis* (CIA; Granger 2015). CIA originally focused on comparisons between learner language and corresponding native-speaker or expert varieties. For example, researchers might compare the frequencies of particular words, grammatical constructions, collocations, or discourse markers in learner essays and expert academic writing.

Most learner corpus research has used a *cross-sectional* design, in which data are collected from different learners at one point in time. Researchers may then compare groups representing different proficiency levels or instructional stages. Cross-sectional designs are relatively practical because they do not require researchers to follow the same individuals for an extended period.

A *longitudinal* design, by contrast, collects repeated samples from the same learners over time. Longitudinal learner corpora are less common because they require sustained access to participants and must address issues such as participant attrition, changes in instructional context, and variation in elicitation tasks. Nevertheless, longitudinal learner corpus research is increasing because it provides more direct evidence of individual development.

A third approach is sometimes described as *pseudo-longitudinal*. In this design, researchers collect data from cohorts of learners at different proficiency levels at approximately the same point in time. The resulting differences do not demonstrate that particular individuals moved through the observed stages. However, they can provide detailed descriptions of different points along an assumed interlanguage continuum. This approach is popular in learner corpus research and can inform the development of teaching materials or assessment resources designed for particular proficiency levels.

2.1.3 Linguistic foci

Learner corpus research investigates a broad range of linguistic phenomena, although some areas have received considerably more attention than others. A survey of the Learner Corpus Bibliography found that (1) lexical phenomena were examined in approximately 65% of studies, followed by (2) grammar at 46%, (3) discourse at 30%, and (4) pragmatics at 10%. Note that these categories are not mutually exclusive, so an individual study may contribute to more than one area.

Within lexical research, phraseology has been particularly prominent. Researchers have examined learners' use of collocations, lexical bundles, formulaic sequences, and associations between words. These studies may investigate whether learners use a narrower range of combinations, rely heavily on a small number of familiar expressions, or produce combinations that are grammatically acceptable but uncommon in the relevant register.

Grammatical learner corpus research includes topics such as tense and aspect, article use, modality, complementation, and syntactic construction use. Discourse and pragmatic studies frequently investigate features such as stance markers, engagement markers, connectors, discourse-organizing expressions, and strategies for expressing politeness or evaluation.

Most learner corpus studies have used written rather than spoken data. Academic writing has received especially strong attention, partly because written texts are easier to collect, store, and annotate, and partly because advanced academic writing creates important pedagogical needs. This emphasis has contributed valuable knowledge about L2 writing, but it also means that conversational language, informal registers, interactional development, and lower-proficiency production remain comparatively underrepresented.

2.1.4 Contributions to SLA

One of the original objectives of learner corpus research was to contribute to SLA theory. For a considerable period, however, the theoretical grounding of LCR remained relatively weak. Many early studies were primarily descriptive, reporting frequencies or differences without clearly connecting those observations to theories of language acquisition. This partly explains the initially limited uptake of learner corpus findings in the broader SLA community.

The situation has changed substantially (see Tracy-Ventura and Paquot, 2021). Learner corpus data now have a more established role in research on L2 acquisition, although corpus-based production data remain less central than experimental data (e.g., comprehension measures, acceptability judgments, reaction times, and online processing data). Learner corpora cannot directly reveal all of the cognitive processes underlying language acquisition. They can, however, provide large and diverse collections of naturally occurring or task-elicited learner production.

The scale and diversity of learner corpus data can make it possible to test whether findings based on a limited sample generalize across learner populations. For example, Murakami and Alexopoulou (2016) used large-scale learner data to challenge claims concerning a universal order of acquisition for L2 English morphemes. Their findings demonstrated that learners' first-language backgrounds substantially influenced the observed acquisition patterns. This illustrates how learner corpus evidence can qualify conclusions that may appear universal when they are based on only a small number of learner groups.

Learner corpus research has also made important contributions to the study of *linguistic complexity*. Corpus-based studies have helped researchers *operationalize* complexity by developing systematic measures of grammatical, syntactic, and lexical complexity. These measures have been used to examine how complexity develops over time and varies according to proficiency, L1, register, genre, and task complexity (e.g., Biber et al., 2020; Kyle et al., 2024; Sung and Kyle, 2025). A closely related area involves *predictive* studies. In addition to describing group-level differences, these studies examine which factors are associated with learners' linguistic choices or learning outcomes. For example, researchers may investigate which lexical, syntactic, phraseological, or morphological features predict human judgments of writing or speaking proficiency. Such analyses can inform theories of L2 development, but correlations between linguistic features and proficiency scores should not be interpreted as direct evidence of acquisition or causal relationships.

2.1.5 Contributions to L2 teaching and assessment

From its beginnings, learner corpus research has been strongly committed to improving L2 teaching through more learner-focused materials and pedagogical practices. Many of these resources have been designed for higher-intermediate or advanced learners, whose difficulties may involve subtle restrictions on usage, register, phraseology, and discourse organization rather than only basic grammatical errors.

One major area of application concerns *accuracy*. Learner corpus data can be used to identify recurrent grammatical errors, lexical infelicities, problematic collocations, and non-target-like phraseological patterns. These findings can then support materials intended to raise learners' awareness of common difficulties and improve grammatical, lexical, and phraseological accuracy. One example is the Louvain English for Academic Purposes Dictionary, or LEAD (Granger and Paquot, 2015, 2022). This web-based academic writing resource draws on analyses of academic texts produced by learners from several L1 backgrounds. It highlights cross-disciplinary academic words that learners frequently misuse, overuse, or underuse. It also presents their phraseological patterning, including common collocations and lexical bundles. The resource demonstrates how learner corpus findings can be translated into pedagogical support that goes beyond simple word definitions.

Learner corpora can also be incorporated into *data-driven learning* (DDL). In DDL activities, learners examine corpus examples and identify linguistic patterns for themselves, often with guidance from the teacher. Learner-corpus-based DDL may expose students to examples of successful learner production, recurrent difficulties, or alternative revisions. Preliminary studies suggest that this form of instruction can be effective and is often received positively by students and teachers (Hartle, 2023). However, the direct use of L2 data in DDL remains relatively limited. More research is needed to determine which types of learner data are most useful, how examples should be selected and presented, and under what conditions learner-corpus-based DDL improves learning.

Learner corpus evidence may also inform proficiency-level descriptions, teaching sequences, and assessment criteria. For example, pseudo-longitudinal analyses can identify features that become more frequent, diverse, accurate, or sophisticated at different proficiency levels. These patterns may contribute to resources associated with frameworks (e.g., Common European Framework of Reference for Languages [CEFR]). However, corpus frequencies should not be converted directly into proficiency descriptors without considering task, register, learner population, and the communicative function of the feature.

2.2 How can LCR be connected to NLP?

At this point, we might have a good understanding that NLP broadly refers to the use of computational methods to analyze human language automatically (not synonymous with artificial intelligence, machine learning, or statistics, although it may draw on these approaches). NLP tools of varying levels of complexity have played an important role in the development of corpus linguistics generally and learner corpus research in particular. Relatively simple tools include concordancers, which retrieve all occurrences of a search word or expression and display them in their surrounding contexts, allowing researchers to identify and compare recurring linguistic patterns. More complex tools, such as part-of-speech (POS) taggers and syntactic parsers, automatically enrich corpus texts with grammatical and syntactic information. The following section first introduces several NLP processes commonly used to annotate and analyze learner corpora.

2.2.1 Common NLP (pre-)processes

Tokenization. Tokenization (i.e.g, the division of a text into word-like units called *tokens*) is typically the very first stage of an NLP pipeline. Although tokenization may appear straightforward in languages in which words are generally separated by spaces, even an English orthographic word may contain multiple linguistically meaningful components. Tokenization becomes more challenging in some languages (e.g., Korean), in which individual space-delimited units may regularly contain multiple morphemes. Decisions about token and morpheme boundaries can therefore become an important source of variation in corpus analysis.

Example: Tokenization in English and Korean

English sentence

She walked to school.

Word-level tokens

She |walked |to |school |. (5 tokens)

Morpheme-level units

she |walk |-ed |to |school |. (6 tokens)

(*Note.* Punctuation counted as a separate token)

Korean sentence

저는	학교에	갑니다.
저 + 는	학교 + 에	가 + ㅂ니다
I + topic marker	*school + locative marker*	*go + declarative ending*

'*I go to school.*'

Space-delimited units

저는 |학교에 |갑니다 |. (4 tokens)

Morpheme-level units

저 |는 |학교 |에 |가 |ㅂ니다 |. (7 tokens)

Lemmatization. Lemmatization reduces inflected word forms to a shared dictionary form or lemma. For example, *walks*, *walked*, and *walking* may be analyzed as forms of the lemma *walk*. Lemmatization is widely used to calculate lexical frequencies and diversity, although it is not necessary or preferable for every language or research question. Combining forms at the lemma level may obscure meaningful morphological variation, particularly when inflectional choices are themselves central to the analysis.

Example: Lemmatization and word-type counts

Example sentence

She *walks* to school. She *walked* home. She enjoys *walking*.

Word-form sequence

she |walks |to |school |she |walked |home |she |enjoys |walking

Lemmatized sequence

she |walk |to |school |she |walk |home |she |enjoy |walk

Count	**Before lemmatization**	**After lemmatization**
Tokens	10	10
Types	8	6

Word-form types:

{*she*, ***walks***, *to*, *school*, ***walked***, *home*, *enjoys*, ***walking***} = 8

Lemma types:

{*she*, ***walk***, *to*, *school*, *home*, *enjoy*} = 6

Tagging. Tagging commonly refers to the assignment of grammatical part-of-speech (POS) labels to words or tokens in a text. POS tagsets can be classified as either *coarse-grained* or *fine-grained*. Coarse-grained tagsets distinguish broad grammatical categories without encoding many distinctions within each category. For example, the Universal POS tagset (UPOS; Nivre et al., 2020) consists of 17 tags representing major word classes, including adjectives (`ADJ`), adverbs (`ADV`), nouns (`NOUN`), and verbs (`VERB`). Although UPOS distinguishes proper nouns from common nouns, it generally provides limited information about grammatical subclasses or inflectional forms within each word class.

Fine-grained tagsets encode more detailed, and often language-specific, distinctions within these broad POS categories. The Penn Treebank tagset (Marcinkiewicz, 1994; Santorini, 1990), for instance, contains 36 POS tags. Whereas UPOS assigns all common nouns the single tag `NOUN`, the Penn Treebank tagset distinguishes singular or mass nouns (`NN`) from plural nouns (`NNS`). The difference is even more apparent for verbs. UPOS generally assigns lexical verbs the single tag `VERB`, while the Penn Treebank tagset distinguishes six inflectional forms: base form (`VB`), past tense (`VBD`), gerund or present participle (`VBG`), past participle (`VBN`), non-third-person singular present (`VBP`), and third-person singular present (`VBZ`). The Sejong Korean tagset is a specialized tag set for tagging Korean morphemes.[5]

Figure 1 illustrates how an example sentence from the English Web Treebank (Bies et al., 2012; Silveira et al., 2014) is represented using the coarse-grained UPOS tagset and the more fine-grained Penn Treebank tagset.

	1	*2*	*3*	*5*	*6*	*8*	*9*	*10*	*11*	*12*	*13*	*14*	*15*
Token	His	artworks	were	selected	and	exhibited	in	the	British	Museum	in	2002	.
UPOS tags	PRON	NOUN	AUX	VERB	CCONJ	VERB	ADP	DET	PROPN	PROPN	ADP	NUM	PUNCT
Penn tags	PRP$	NNS	VBD	VBN	CC	VBN	IN	DT	NNP	NNP	IN	CD	.

Figure 1: A comparison of UPOS tags vs. Penn Treebank tags. Adapted from Kyle and Sung (2026)

[5]See examples here: `https://nlpxl2korean.github.io/UD-KSL/xpos/`

Parsing. *Parsing* refers to the automatic analysis and annotation of syntactic structure in a text. Two major approaches are commonly used: (1) *constituency parsing* and (2) *dependency parsing*. Although both approaches represent relationships among words and larger syntactic units, they differ in how sentence structure is organized and described.

(1) A *constituency parse* represents a sentence or utterance as a hierarchical tree composed of nested phrases and clauses. Words are grouped into syntactic constituents (e.g., noun phrase, verb phrases), which are in turn combined into larger structures. Figure 2 presents an example of a constituency parse tree. Although several constituency-based annotation frameworks exist, the Penn Treebank syntactic annotation scheme has been particularly influential in English-language treebanks,[6] including the Penn Treebank (Marcinkiewicz, 1994) and OntoNotes (Weischedel et al., 2022). The Penn Treebank scheme includes labels for major phrase and clause categories, such as declarative clauses (`S`), noun phrases (`NP`), and verb phrases (`VP`). These labels represent the hierarchical organization of syntactic constituents within a sentence.

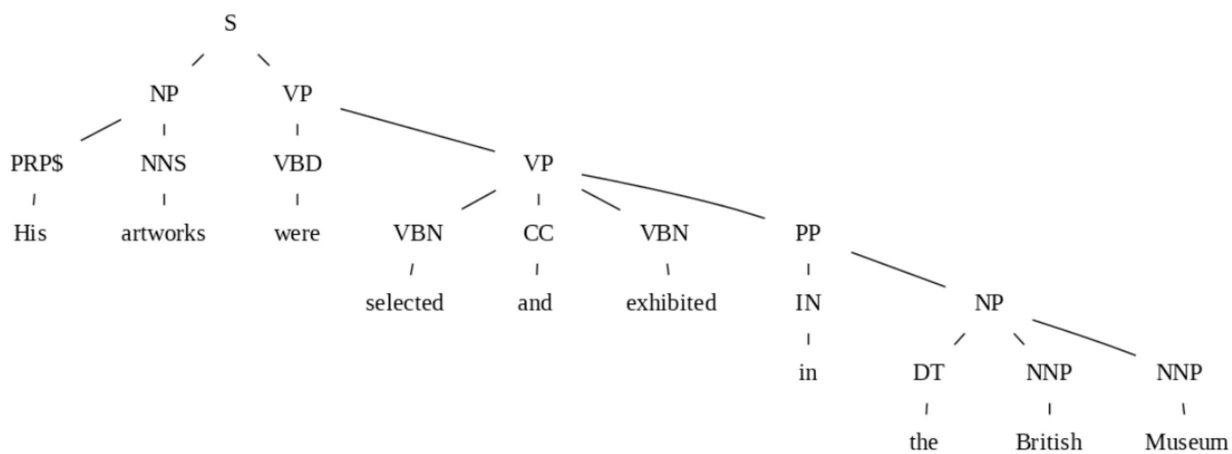

Figure 2: An example of constituency parse tree, visualized using *spaCy* with the *benepar_en3 parser*. Adapted from Kyle and Sung (2026)

(2) A *dependency parse* represents syntactic structure as a set of directed relationships between individual words. Each relationship connects a syntactic *head* to one of its *dependents* and is assigned a label indicating the grammatical function of that relationship. Figure 3 shows an example of a dependency parse.

[6] A *treebank* is a corpus in which sentences have been annotated with syntactic structure.

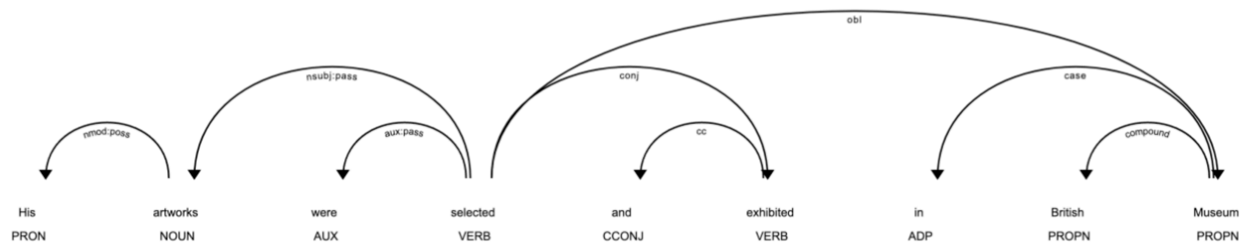

Figure 3: An example of dependency parse tree with Universal Dependencies, visualized by *displaCy*. Adapted from Kyle and Sung (2026)

Some dependency annotation schemes are designed for particular languages, such as CLEARNLP for English (Choi et al., 2015), whereas others are intended to support consistent annotation across many languages. Universal Dependencies (UD), for example, provides a cross-linguistically applicable framework for representing grammatical relationships (Nivre et al., 2020).

Compared with constituency representations, dependency representations explicitly encode grammatical functions between words. For example, UD distinguishes several types of nominal dependents associated with predicates, including nominal subjects (`nsubj`), objects (`obj`), and oblique nominals (`obl`). It also distinguishes relationships between nouns, such as nominal modifiers (`nmod`) and appositional modifiers (`appos`). These dependency labels make it possible to identify not only which words are syntactically related, but also the grammatical function of each relationship.

2.2.2 Challenges in evaluating NLP accuracy

One of the challenges associated with these NLP procedures is that automatic annotation is usually produced through a *pipeline* of interdependent processes. Sentence segmentation may be followed by tokenization, lemmatization, tagging, and parsing, such that errors introduced at an early stage can propagate to subsequent stages and affect the final annotation. However, published accuracy estimates often evaluate individual components in isolation while assuming that all preceding steps have been completed perfectly. For example, POS-tagging accuracy may be calculated using manually corrected token boundaries rather than the automatically generated tokenization that researchers would encounter in actual applications.

NLP systems are commonly evaluated by training an algorithm on approximately 90–95% of a manually annotated corpus and testing it on the remaining 5–10%. During this evaluation, the automatic output must be aligned with the manually annotated reference data. Evaluating a single stage independently is useful for comparing systems and improving individual components, but it can create misleading expectations about performance in a complete learner corpus pipeline.

Overall accuracy may also conceal substantial variation across linguistic features. Frequent or relatively straightforward categories may approach perfect accuracy, while rare or ambiguous categories remain poorly annotated. A single overall accuracy score therefore does not necessarily indicate whether the specific linguistic feature examined in a study has been annotated reliably.

A further issue is that NLP tools are not necessarily designed to identify the linguistic distinctions of greatest interest to SLA researchers (Meurers and Dickinson, 2017). A general-purpose parser may perform well according to its original evaluation criteria while still requiring additional processing to identify a particular construction or complexity feature. Researchers must therefore evaluate not only the general performance of a tool but also the accuracy of the specific output on which their analysis depends. This evaluation is difficult because manually annotated L2 corpora suitable for training and testing remain limited. Without appropriate reference data, researchers may have little evidence regarding how accurately a tool processes the learner population, genre, proficiency level, or linguistic feature under investigation. The followings are two illustrative studies that evaluated NLP preprocessing pipelines for L2 English (Kyle et al., 2024) and L2 Korean (Sung and Shin, 2025).

Chapter Summary

Learner corpus research uses collected samples of learner language to investigate the characteristics and development of second language use. It can contribute both to SLA research, by describing variation across learners, tasks, proficiency levels, and learning contexts, and to language teaching and assessment. Learner corpus studies may use cross-sectional, longitudinal, or pseudo-longitudinal designs and may focus on lexical, grammatical, phraseological, discourse, pragmatic, or interactional features. NLP supports learner corpus research by automating processes such as tokenization, tagging, and parsing. These processes make it possible to annotate and analyze datasets that may be too large for entirely manual analysis. However, NLP tools are typically developed and evaluated on particular languages, genres, populations, and annotation schemes. Their reported overall accuracy may therefore provide limited evidence about their performance on learner language or on the specific linguistic feature required for a study. Because errors can propagate through an NLP pipeline, researchers should evaluate the accuracy of the particular annotations, categories, or measures on which their conclusions depend.

Activity: Connecting NLP to your RQ

1. Revisit the RQ you are considering for your final project. What would you like to understand about learner language, language development, language teaching, or assessment?
2. What type of learner language data could help you investigate this question? For example, would you examine learner writing, spoken interviews, classroom interaction, assessment responses, discussion forums, or repeated samples collected over time?
3. What research design would be most appropriate: cross-sectional, longitudinal, or pseudo-longitudinal? What conclusions would this design allow you to make, and what conclusions would it not support?
4. What information about the learners, tasks, texts, or learning contexts would be necessary to interpret the corpus appropriately? Consider variables such as first-language background, proficiency, age, instructional context, genre, topic, and task conditions.
5. What linguistic feature or construct would you examine? For example, you might investigate vocabulary, phraseology, grammatical constructions, errors, syntactic complexity, discourse organization, pragmatic functions, or interactional features. How would you define and operationalize this construct?
6. What NLP process or tool could support the analysis?
7. How would you determine whether the NLP output is sufficiently accurate for your research question?
8. What part of the analysis would still require human judgment? Consider tasks such as identifying learner errors, interpreting intended meanings, resolving ambiguous annotations, or evaluating whether a computational measure represents the intended construct.

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